

The Sentinel Firewall: a model-based economic evaluation of safe-product substitution to reduce lead exposure in LMICs

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Abstract

Introduction

Lead exposure remains an important and under-addressed driver of neurodevelopmental and cardiovascular inequity in low- and middle-income countries (LMICs). Much of the remaining burden now comes from informal consumer products that households cannot easily identify and regulators do not easily control, including artisanal cookware, traditional cosmetics and lead-contaminated food-contact items. We evaluated a Sentinel Firewall strategy in which high-risk districts are identified through sentinel screening and safer substitute products are delivered through maternal and child health contacts.

Methods

We developed a model-based economic evaluation comparing two safe-product packages against current practice without Sentinel Firewall-based substitution. The Kitchen Package included a lead-safe pot voucher, calcium and safe child utensils. The Kohl Package included safe maternal and infant kohl substitutes. From a social perspective, the model incorporated district fixed costs, variable programme costs, maternal-fetal transfer of lead, developmental timing of lead reduction, child cognitive and lifetime earnings gains, maternal and neonatal health gains, and adult cardiovascular gains as a secondary extension. Intervention costs were modelled in 2026 US dollars, while lifetime earnings were valued using a purchasing-power-parity productivity anchor equal to GDP per capita multiplied by labour's share of national income. Uncertainty was characterised with 10 000 Monte Carlo draws using PERT distributions and modest positive dependence across key implementation steps.

Results

In the baseline specification, the Kitchen Package generated a median cognition-only benefit-cost ratio of 4.61 (5th-95th percentile 1.65 to 12.79), and the Kohl Package generated a median cognition-only benefit-cost ratio of 6.24 (1.96 to 17.98). Adding

maternal and neonatal health gains increased the median benefit-cost ratios to 4.80 for the Kitchen Package and 6.55 for the Kohl Package. Adding adult cardiovascular gains increased the median total benefit-cost ratios to 5.69 and 6.82, respectively. In both packages, the 5th percentile of the total social benefit-cost ratio remained above parity.

Conclusion

Targeted safe-product substitution through maternal and child health platforms appears socially cost-effective in hotspot settings where hazardous consumer products are common. The model is intended as a decision tool rather than as a universal estimate: the public spreadsheet and Python code allow ministries, NGOs and local researchers to replace the illustrative assumptions with local data. These findings support targeted pilot implementation of safe-product substitution while stronger regulatory and industrial reforms are pursued.

Summary box

What is already known

- Lead exposure remains a major cause of neurodevelopmental loss and adult cardiovascular disease in LMICs.
- Important contemporary exposure routes often arise from informal consumer products such as artisanal metal cookware, traditional leaded cosmetics and lead-contaminated food-contact items.
- Pregnancy, infancy and early childhood are especially vulnerable windows because maternal lead can be mobilised during pregnancy and transferred to the fetus, while infants and young children absorb more ingested lead than adults.

What are the new findings

- We propose a Sentinel Firewall framework that identifies high-risk districts or catchments and uses existing maternal and child health platforms to distribute safe-product substitutes targeted to locally identified hazards.
- We evaluate a Kitchen Package delivered through antenatal care and immunisation contacts, and a Kohl Package delivered through antenatal care plus either facility delivery or early-immunisation catch-up.
- In a 10 000-draw Monte Carlo simulation, both packages have median benefit-cost ratios above parity. The lower-cost Kohl Package has the higher cognition-only median BCR, while the Kitchen Package delivers larger absolute child BLL and earnings gains and the tighter lower tail once broader benefits are included.
- The model is designed as a transparent local-adaptation tool: users can replace the illustrative LMIC hotspot assumptions with local prevalence, cost, uptake and blood lead response data.

What do the new findings imply

- Health ministries may be able to reduce lead exposure during the highest-risk life stages by integrating targeted safe-product replacement into existing maternal and child health contacts.
- Sentinel screening and voucher-linked substitution can provide immediate risk reduction while generating intelligence about where to target enforcement and producer upgrading.
- Because almost all parameters remain uncertain, pilot programmes should prioritise measurement of product prevalence, redemption, sustained use, residual unsafe use and measured blood lead response.
- The spreadsheet and Python code are intended to help ministries, NGOs and local researchers test whether substitution appears cost-effective in their own regions before scaling.

Introduction

Lead exposure in LMICs increasingly reflects a regulatory mismatch. Although the long-term goal should be to remove lead from the global economy, millions of tonnes of lead continue to circulate through mining, manufacturing, recycling and informal supply chains, with harms falling disproportionately on LMICs.¹ After the phase-out of leaded gasoline, a larger share of the remaining burden now comes from the "last mile" of lead exposure: ordinary consumer products that households use every day, but that regulators do not easily observe or control. These include artisanal cookware cast from mixed scrap, traditional eye cosmetics such as kohl or surma, and ceramics or other food-contact goods containing lead. WHO identifies cosmetics, ceramic glazes, toys, traditional medicines, paint and contaminated food or water as continuing sources of lead exposure, while recent rapid market screening across 25 LMICs found frequent contamination in metal foodware, ceramics, paints and cosmetics.^{2,3}

These products may be widely trusted, culturally embedded and difficult for households to distinguish from safer alternatives. Because lead toxicity is almost always invisible in the short run, households receive no immediate warning signal that would trigger avoidance. But low visibility does not imply low importance: WHO estimates that lead exposure caused more than 1.5 million deaths in 2021, primarily through cardiovascular disease, and substantial disability worldwide.²

The regulatory mismatch matters especially during pregnancy and early childhood. Lead stored in bone can be released into blood during pregnancy and become a source of fetal exposure. Infants and young children are also more vulnerable because they absorb more ingested lead than adults and because the developing brain is highly sensitive to toxic injury.² These overlapping pathways mean that a practical lead-control strategy in many LMIC settings must do two things at once: identify local exposure clusters and protect households during the life stages when developmental harm is largest.

Three tractable exposure pathways

The relative importance of lead sources varies across regions, but three consumer-product pathways illustrate a broader pattern of informally produced, weakly regulated goods that can generate large exposure in specific communities.

First, artisanal aluminium cookware can leach lead into food when pots are cast from mixed scrap containing leaded materials. Studies of metal cookware in Cameroon and other settings show that lead can leach from artisanal aluminium pots at levels capable of producing substantial dietary exposure.^{4,5} Second, lead-contaminated cups, mugs, bowls and spoons can increase chronic ingestion among infants and toddlers once complementary feeding begins (that is, when foods and liquids are introduced in addition to breast milk). The parameterisation in this paper treats this pathway as plausible but less directly measured than the cookware pathway, drawing on the broader literature on lead-contaminated foodware and lead-glazed ceramics.^{3,6} Third, traditional eye cosmetics such as kohl, kajal or surma often begin in the neonatal period for perceived protective, ritual or health reasons. These products can contain very high lead concentrations, and clinical and surveillance reports document exposure among mothers and children.^{3,7}

The three sources are attractive targets for policy because substitutes exist. Unlike diffuse ambient contamination, contaminated cookware, foodware and kohl can in principle be replaced quickly if safe alternatives are culturally acceptable and governments can create credible supply chains for safe products.

Pregnancy as a window of vulnerability and opportunity

Pregnancy is both a biological vulnerability and a programmatic opportunity. Lead stored in maternal bone can be mobilised during pregnancy and lactation, and maternal blood lead is strongly related to fetal exposure. Nutritional deficits, especially low calcium intake, can worsen this process by increasing mobilisation or absorption of lead; calcium supplementation during pregnancy has been shown in a randomised trial to reduce maternal blood lead levels.^{2,8} Antenatal care (ANC) reaches a large share of pregnant women in many LMICs, and health systems already use ANC contacts to provide counselling, supplements and referrals.⁹ Routine immunisation visits create an additional platform for reaching infants when complementary feeding begins and child exposure pathways expand.^{10,11} Facility delivery is another possible contact for newborn interventions, but delivery and immunisation contacts are correlated rather than independent. The model therefore treats infant kohl delivery as occurring through the union of facility delivery or early-immunisation catch-up.

The Sentinel Firewall framework

We propose a Sentinel Firewall framework in which maternal and child health services act as a temporary consumer-protection layer while environmental enforcement and producer upgrading catch up. First, districts or subdistricts with evidence of concentrated product

contamination are identified through sentinel screening, including market sweeps, rapid field screening of high-risk goods, clinic-linked intelligence and, where feasible, sentinel child blood lead monitoring. Rapid market screening with X-ray fluorescence has already been used across 25 LMICs to identify contaminated consumer products and foodware categories that could plausibly guide local prioritisation.³

Second, when a district or catchment is flagged, health contacts trigger modular intervention packages targeted to the locally dominant exposure routes. Third, the programme closes the loop by feeding information from product replacement and biomonitoring back into enforcement. If sentinel sweeps, voucher redemption or surrendered products repeatedly identify specific hazardous goods or supply chains, those data can help prioritise enforcement and producer conversion to safe products.

The Sentinel Firewall is not a substitute for structural lead control. It is an emergency harm-reduction strategy designed to buy time, protect the most vulnerable groups and generate actionable intelligence to reduce the supply of products containing lead.

The safe-products packages

We evaluate two intervention packages. The Kitchen Package includes a voucher distributed through ANC for a lead-safe cooking pot, calcium delivered through ANC and safe child utensils delivered at infant immunisation visits. The package is designed to protect the pregnant woman, reduce fetal exposure through lower maternal blood lead and lower direct child exposure after birth.

The Kohl Package includes safe maternal kohl delivered through ANC and safe infant kohl delivered through either facility delivery or early-immunisation catch-up. The main model allows infant kohl exposure among boys as well as girls, because clinical evidence and programme logic suggest that neonatal and first-year use can affect both sexes even where use persists longer among girls in some settings.⁷ A robustness check restricts infant kohl substitution to girls only. The prevalence parameters for kohl should therefore be interpreted as conditional on screened hotspot settings rather than as unconditional regional averages.

Methods

This report follows the CHEERS 2022 reporting framework for health economic evaluations as applied to a model-based policy analysis.³⁷ No prespecified health economic analysis plan was registered; the model structure, parameter ranges and reporting choices were developed iteratively from the literature and then documented in the public code, spreadsheet and appendices.

Study design, decision problem and comparator

The model asks a simple policy question: if a health system uses maternal and child health contacts to replace leaded products with safer substitutes in high-risk districts, are the

resulting social gains likely to exceed programme costs? The paper has two linked contributions. First, it demonstrates what follows when plausible LMIC values are applied to settings where hazardous cookware, kohl or child feeding items are common. Second, it provides a transparent tool that local analysts can adapt with their own prevalence, cost, uptake and blood-lead data. The comparator is current practice without a Sentinel Firewall programme of safe-product substitution. The purpose is therefore not to estimate one global benefit-cost ratio for all LMICs, but to make the assumptions and uncertainty structure explicit enough for local decision-making. We answer that question by linking each package to changes in blood lead, then translating those reductions into child cognitive gains, maternal and neonatal health gains, and adult cardiovascular gains (see Figure 1 for the main cognitive pathway).

Figure 1A. Core causal pathway

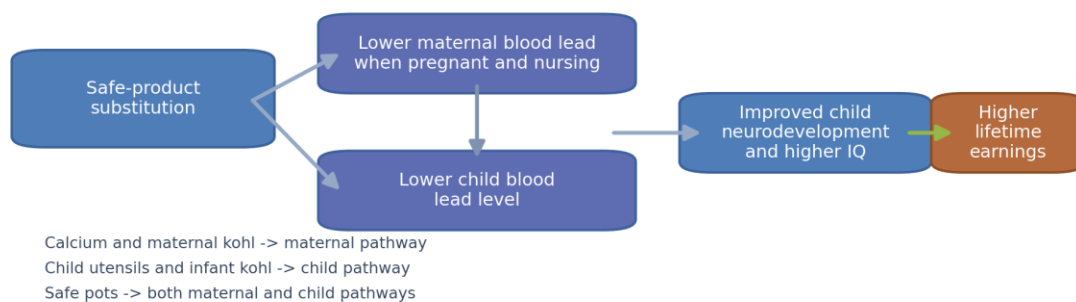
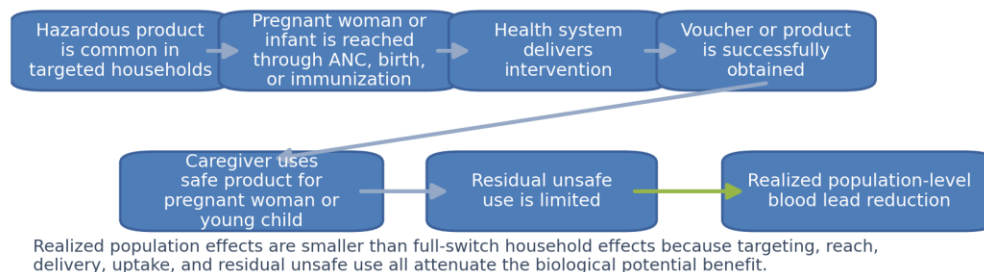


Figure 1B. Implementation chain



Note: IQ gains depend on the size, duration and developmental timing of fetal and child blood lead level (BLL) reductions. Safe pots and calcium can lower maternal BLL during pregnancy and lactation, reducing fetal and breastmilk-mediated infant exposure. Safe pots and child utensils can also reduce postnatal exposure once complementary feeding begins.

Setting, perspective, time horizon and currency

The target setting is a sentinel-identified high-risk district or catchment in an LMIC in which one or more hazardous consumer products are common. The target population is the mother-child pair reached through antenatal care, facility delivery, and early childhood

immunisation contacts, with secondary adult cardiovascular effects allowed for mothers and, in the Kitchen Package, probabilistically weighted co-resident grandparents. The primary analysis takes a social perspective. Costs are modelled per mother-child pair reached and are reported in 2026 US dollars. Earnings are valued from birth over the working life, while maternal, neonatal and adult cardiovascular effects are discounted to the time of intervention or birth according to the relevant pathway. Intervention-cost inputs are expressed directly in 2026 US dollars. Lifetime productivity is valued separately using the PPP GDP anchor described below rather than by converting wages from another currency-year dataset.

Model structure and rationale

We use a probabilistic Monte Carlo model because the intervention is a policy package rather than a single biological treatment effect, and the key quantities of interest depend on implementation losses, duration of protection, developmental timing and multiple uncertain valuation parameters. The model has a mother-child pair as its primary analytic unit. It follows an implementation chain from targeting and health contact through delivery, redemption or receipt, use of the safe product, and residual unsafe use. It then maps the resulting maternal and child blood lead reductions into child cognition and lifetime earnings, maternal and neonatal health gains, and adult cardiovascular gains.

Measurement and valuation of outcomes

The primary economic channel is child cognition. For each simulated mother-child pair, we estimate the present value of lifetime earnings from birth using a market-labour productivity anchor equal to purchasing-power-parity (PPP) GDP per capita multiplied by labour's share of national income, combined with survival to adulthood and the end of working life, expected real income growth and a social discount rate, that is, the rate used to convert future benefits into present-value terms. This follows the social benefit-cost tradition of valuing cognitive gains at their marginal product rather than at observed wage rates, which would miss own-account and informal production that remain important in the settings modelled here. Household production, subsistence output not captured in national accounts and the intrinsic value of cognition beyond earnings are excluded, making this earnings channel conservative on those dimensions.¹²⁻¹⁴ To reflect wide uncertainty across plausible LMIC settings, these macroeconomic inputs are assigned ranges rather than point values: PPP GDP per capita ranges from US\$2000 to US\$8000 with a mode of US\$4000, the labour-share parameter ranges from 0.50 to 0.58 with a mode of 0.55, real productivity growth has a mode of 2%, and the social discount rate has a mode of 3%.¹²⁻¹⁴

We convert child blood lead reduction into IQ gain using a linear parameter of 0.20-0.60 IQ points per 1 ug/dL reduction in developmentally averaged child blood lead, with a mode of 0.35. This is a practical approximation to a nonlinear literature showing cognitive harm at blood lead levels below 10 ug/dL and larger marginal harm at lower levels.¹⁵⁻¹⁸ A recent

LMIC learning synthesis provides an additional check that lead exposure is meaningfully associated with lower learning outcomes in developing-country settings.¹⁹

We map IQ gains into lifetime earnings using an LMIC-centred return of 0.0030-0.0100 lifetime-earnings units per unit of baseline lifetime earnings for each IQ point gained, with a mode of 0.0050. In other words, the mode assumes that one additional IQ point raises discounted lifetime earnings by about 0.5%. This range is intended to be policy-conservative: the lower bound allows for substantial labour-market friction in subsistence-heavy settings, the mode uses the lower end of lead valuation estimates, and the upper bound is aligned with developing-country cognitive-skill estimates while avoiding higher US-based values.²⁰⁻²³

Data sources and key assumptions

The model distinguishes direct child lead reductions from maternal pathways. In the Kitchen Package, the safe pot and safe utensils reduce direct child exposure, while the safe pot and calcium reduce maternal blood lead during pregnancy; persistent maternal pot use can also reduce infant exposure during the milk-dominated part of the first year. In the Kohl Package, maternal kohl affects prenatal exposure and can also lower early-infancy exposure through the same maternal pathway, while infant kohl affects direct child exposure after birth. To reflect placental transfer, maternal blood lead reductions during pregnancy are converted into fetal lead reductions using a fetal-transfer parameter with a mode of 0.80. The parameter is motivated by the strong relationship between maternal and cord blood lead reported across paired-sample studies and by evidence that reducing maternal blood lead during pregnancy can lower fetal exposure.^{2,8,24} In paired maternal-cord studies, cord blood lead is often a large fraction of concurrent maternal blood lead, commonly on the order of 70%-100%, which supports the model's modal fetal transfer coefficient of 0.80 as a structural approximation rather than as a directly estimated child-IQ coefficient.^{2,24}

The cognition model does not treat a short prenatal or one-year infant intervention as if it reduced average blood lead by the same amount across all formative years. Instead, it assigns explicit developmental shares of total cognition-relevant harm that sum to one: 20%-40% of cognitive harm in the last four months in utero, with a mode of 30%; 20%-40% in year 1, with a mode of 30%; and the remaining 30%-60% in years 2-5, with a mode of 40%. This timing adjustment is motivated by evidence linking cognition to prenatal exposure, exposure during infancy, and repeated early-childhood exposure histories rather than to a single brief exposure interval.^{17,18,26}

Within that structure, direct child cookware and utensil pathways begin with complementary feeding rather than at birth, whereas persistent maternal pot use or maternal kohl use can also lower infant exposure during the milk-dominated part of the first year. Thus, in the current model, the safe-pot pathway affects the child during 0-6 months only through the maternal lactational channel, while the kohl package has both a maternal lactational channel and a separate direct infant-kohl channel in that same

window. Appendix Table A1 and Appendix Figure A5 summarize these timing windows by intervention component.

The most influential biological efficacy inputs are deliberately assigned wide ranges when the literature is indirect. Among households that use hazardous cookware and then switch fully to the safe pot for food consumed by the pregnant woman or young child, the postnatal child blood lead level (BLL) reduction is modelled as 2-8 ug/dL, with a mode of 5. A full switch means that the household starts with the unsafe product, receives or redeems the safer substitute, and uses the safe product for the target user without enough continued unsafe-product use to preserve the original exposure pathway. The cookware child parameter is defined as the postnatal child effect only; any prenatal child benefit from lower maternal lead is modelled separately through the maternal pot pathway and fetal transfer. The calcium parameter is anchored more tightly because it is informed by a randomised trial in pregnancy: 0.2-1.2 ug/dL, with a mode of 0.4, which is close to the full-sample average reduction reported by Ettinger et al. The upper tail allows for larger effects in women with higher compliance or higher bone-lead burden, but the central value is now aligned to the trial-wide intent-to-treat result rather than to a subgroup effect.⁸

We model multiple imperfections in implementation. For calcium, maternal kohl, infant kohl and child utensils, realised benefit depends on baseline use of the hazardous product where relevant, attendance at the relevant health contact, fidelity of the health system in actually delivering both the safe product and the intended counselling once that contact occurs, and sustained use. For the pot, we model a longer voucher chain in which the health care system must issue the voucher, a participating merchant must have the safe pot in stock when the voucher is presented, the woman must redeem the voucher and the new pot (but not the old) must then be used for food consumed by the pregnant woman or toddler. Conditional on redeeming a voucher for a valuable durable good, use by the pregnant woman or toddler may be high because redemption itself is effortful and self-targeting.²⁷ The main concern is therefore non-exclusive use rather than simple non-use: households may continue using the old unsafe pot for some dishes or some family members even after redemption. Calcium is different because the product is clinically familiar but adherence can be undermined by pill burden, side effects, stock interruptions, or weak counselling. Kohl is different again because adherence depends more strongly on local culture and household authority: some caregivers may welcome a safer substitute, while others may continue to trust traditional galena-based products. We therefore include a residual-use penalty for the pot and retain broad implementation uncertainty for the other components.

Multiplying modal values for targeting, health-care contact, delivery fidelity, household adherence, voucher redemption and residual unsafe use gives a simple sense of the implementation attenuation we assume. The modal assumptions imply that only a small share of the full-switcher biological benefit is realised at the population level after baseline hazard prevalence, contact coverage, product delivery, merchant stock availability, redemption and adherence are all taken into account: approximately 0.20 for the pot pathway, 0.14 for child utensils, 0.15 for maternal kohl and 0.12 for infant kohl. These

attenuations arise from realistic losses at each step of the chain, not from any assumption that the products are biologically ineffective once consistently used.

Costing methods

Costs are tied to actual programme operations. Total programme cost per mother-child pair includes a fully loaded share of national and district-level fixed programme overhead and setup cost, modelled as fixed cost per birth = district fixed cost/annual births in the screening unit. We use a modal effective birth cohort of 20 000 births per screening unit. This reflects a shared-service model in which one screening unit's X-ray fluorescence equipment, training, supervision and market-screening capacity can be spread across multiple smaller districts rather than purchased separately by each district. Variable costs include counselling time, distribution costs, product costs and a programme markup. Product and distribution costs fall when attendance or delivery fails, but fixed surveillance and setup costs remain.

Broader health consequences

The primary outcome for decision-making is the cognition-only benefit-cost ratio. We also report a benefit-cost ratio including maternal and neonatal health gains and a total benefit-cost ratio that further includes adult cardiovascular effects. Our primary economic endpoint is therefore the present value of lifetime earnings preserved through higher child cognitive ability, but lead exposure also affects maternal health, neonatal outcomes and adult cardiovascular disease.

We model maternal and neonatal benefits through reduced pre-eclampsia and preterm birth risk. These pathways operate through reductions in maternal blood lead during pregnancy. The model starts with baseline prevalence values for pre-eclampsia and preterm birth and applies odds ratios per 1 ug/dL lead increase to estimate how much those risks fall when maternal blood lead is reduced. The pre-eclampsia dose-response is closer to a directly reported per-ug/dL estimate than the preterm relationship, for which the literature is broader and includes meta-analytic birth-outcome evidence beyond a single transportable slope.²⁵ Because the intervention begins during pregnancy rather than before conception, the model scales maternal BLL reduction by timing multipliers before applying the pre-eclampsia and preterm equations. The resulting risk changes are converted into monetised health gains using reduced-form disability-adjusted life-year (DALY)-based parameters.²⁸⁻³³

We also model adult cardiovascular benefits as a secondary extension rather than as part of the core child-cognition case. In the Kitchen Package, safer cookware can reduce exposure for the mother and, where relevant, for co-resident grandparents. In the Kohl Package, reduced maternal exposure from safe kohl affects the mother directly. We value these gains using a DALY-based approach that combines an assumed lifetime cardiovascular burden per sustained 1 ug/dL lead reduction with a value of a statistical life year. Because cardiovascular benefits occur later in life, they are discounted according to the assumed age of the exposed adult at the time of intervention.^{34,35} This cardiovascular

term should be read cautiously. The Larsen and Sanchez-Triana study is a single global health-impact and economic-modeling analysis anchored in lifetime lead exposure, whereas our interventions last only a few months or years. We therefore scale the implied lifetime cardiovascular burden by the duration of the modeled exposure reduction as a share of the adult's lifetime. That may be optimistic if short-term interventions do not meaningfully change long-run cardiovascular risk. But it may also be pessimistic if some households continue the safer behavior after the formal intervention ends, so that the true exposure reduction lasts longer than the modeled program duration.

Characterising uncertainty and heterogeneity

Almost all parameters in this model remain uncertain, including product-specific blood lead reductions, implementation probabilities, economic parameters and valuation of non-earnings health gains. We therefore model uncertainty explicitly rather than presenting the analysis as if it depended on a single preferred parameter set.

For each uncertain parameter, we specify a minimum, modal and maximum value and draw from a PERT distribution. This choice reflects the fact that many inputs are only loosely bounded by the literature: we usually have a preferred central estimate, but we do not observe a full empirical sampling distribution for the policy context being modelled. The simulation also includes modest positive dependence across several implementation steps. In practice, attendance, product delivery, voucher issuance, merchant stock availability at the participating retailer, redemption and post-redemption use are unlikely to fail independently. Some districts or facilities will perform better across several links, while others will perform worse.

The Python code exports trial-level data, summary tables, validation checks and figures. Figure 2 compares the Kitchen and Kohl Package distributions of benefit-cost ratios (BCRs). Supplementary figures in the online appendix show package-specific BCR distributions and tornado diagrams identifying the parameters most strongly associated with cognition-only BCRs.

Formal subgroup heterogeneity is limited in the current model. The principal sources of heterogeneity are package-specific implementation chains, developmental timing, stochastic sex assignment for the girls-only infant-kohl robustness check, and probabilistically weighted grandparent co-residence in the adult cardiovascular extension. We did not apply formal distributional weights or inequality aversion parameters. Instead, the equity orientation of the model comes from the intervention design itself: targeted delivery to hotspot districts and to pregnant women, fetuses, infants and young children who face the highest developmental risk from lead exposure.

Results

The Kitchen Package is favourable in the baseline specification, though with substantial uncertainty. The median earnings-only benefit-cost ratio (BCR_Cog) was 4.61, with a 5th to 95th percentile range of 1.65 to 12.79; 0.67% of simulated runs fell below parity.

The baseline Kitchen Package cost a median of US\$15.22 per mother-child pair. It reduced child blood lead by a median of 1.55 ug/dL across pathways during the time the interventions were in use. The population-level reduction is much lower than the modelled biological effect for a household that starts with unsafe products and then switches fully to safe replacements. The difference reflects imperfect targeting, missed health contacts, imperfect health-system distribution, incomplete adherence and residual unsafe-product use. In addition, most new products affect only part of the developmentally important period from late pregnancy through early childhood. After applying both implementation and developmental-timing adjustments, the median effective reduction used for IQ mapping was 0.60 ug/dL. The median IQ gain was 0.21 points, and the median lifetime earnings gain was US\$70.16.

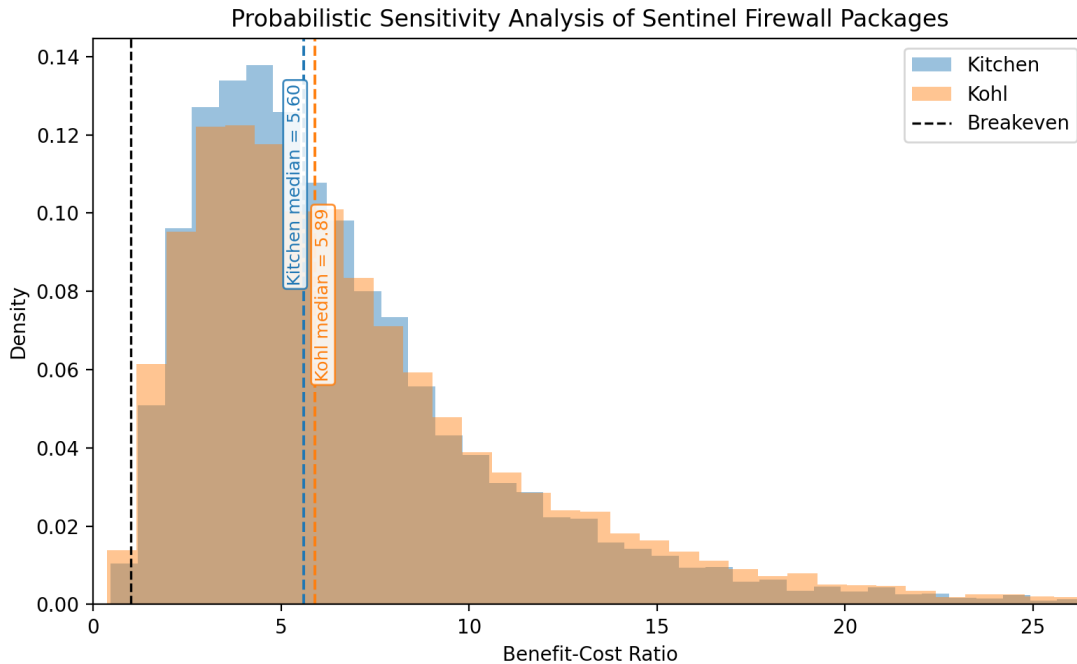
For policy scale, the deterministic modal run implies that in a country with 1 million annual births in sentinel-identified hotspot districts, the Kitchen Package would cost about US\$15.5 million and generate about US\$84.2 million in discounted total social benefits, or about US\$68.7 million in discounted net social benefits.

When maternal and neonatal benefits were added, the median BCR rose to 4.80, with a 5th to 95th percentile range of 1.78 to 13.07; 0.50% of runs fell below parity. Adding adult cardiovascular benefits as a secondary extension increased the median total benefit-cost ratio to 5.69, with a 5th to 95th percentile range of 2.21 to 14.67; 0.09% of runs fell below parity.

The Kohl Package also appears socially cost-effective. In the baseline analysis incorporating only the cognitive pathway, its lower cost produced the higher median benefit-cost ratio: 6.24, with a 5th to 95th percentile range of 1.96 to 17.98; 0.51% of runs fell below parity.

The baseline Kohl Package cost a median of US\$5.75 per mother-child pair. It reduced child blood lead by a median of 1.09 ug/dL during the time the intervention was in use. Because maternal and infant kohl substitution affects a shorter portion of the developmental window, and because the population estimate includes imperfect targeting and delivery frictions, the developmentally weighted reduction used for IQ mapping was 0.31 ug/dL. The median IQ gain was 0.11 points. The corresponding median present value of lifetime earnings gain was US\$36.66.

Under the same deterministic modal scaling, 1 million annual births in sentinel-identified hotspot districts would imply about US\$5.6 million in programme cost, about US\$42.6 million in discounted total social benefits, and about US\$37.1 million in discounted net social benefits for the Kohl Package.



Including maternal and neonatal benefits increased the median benefit-cost ratio to 6.55, with a 5th to 95th percentile range of 2.13 to 18.40; 0.36% of runs fell below parity. Adding adult cardiovascular benefits as a secondary extension increased the median total benefit-cost ratio to 6.82, with a 5th to 95th percentile range of 2.24 to 18.92; 0.30% of runs fell below parity.

Robustness checks

We examined five robustness checks, including several multi-parameter scenarios in which related assumptions shift together. First, the baseline cardiovascular disease (CVD) specification assumes that adult health gains come only from directly modelled target users: the mother and, in the Kitchen Package, a probabilistically weighted co-resident grandparent. If half of recipient households extend safe-product substitution to one additional adult household member, the mode-based adult CVD benefit rises by about 23% in the Kitchen Package and by about 50% in the Kohl Package.

Second, we re-estimated the cognition pathway using a nonlinear log-BLL dose-response. Under a Crump-style specification with $\beta = 3.246$, reducing BLL from 12 to 7 $\mu\text{g/dL}$ implies an IQ gain of about 1.58 points, whereas reducing BLL from 7 to 2 $\mu\text{g/dL}$ implies about 3.18 points. The nonlinear specification therefore gives larger gains when the intervention removes a larger share of total lead exposure and moves children into the lower-BLL range where marginal cognitive harm appears steeper.

Third, we examined alternative background exposure settings. Holding package mechanics at modal values, a low-background scenario with other-source BLL centred at roughly 3 $\mu\text{g/dL}$ (SD 1.5) implies median nonlinear IQ gains of about 0.51 for the Kitchen Package and

0.30 for the Kohl Package. A higher-background scenario with other-source BLL centred at roughly 7 ug/dL (SD 3.5) reduces those median gains to about 0.27 and 0.16, respectively. The corresponding cognition-channel BCRs are about 10.33 for the Kitchen Package and 14.31 for the Kohl Package in the lower-background setting, versus about 5.50 and 7.47 in the higher-background setting. In other words, the economic case is strongest where unsafe cookware or kohl account for a large share of total exposure and weaker where many other sources remain uncontrolled.

Fourth, we model a disadvantaged implementation setting with lower GDP per capita, lower expected productivity growth, lower ANC attendance, weaker delivery fidelity, lower merchant stock availability, lower voucher redemption and lower adherence. Operationally, we reduce the minimum, modal and maximum values of those macroeconomic and implementation parameters by 30%, while leaving biological effect parameters unchanged. As expected, correlated deterioration across these stages substantially lowers the median total BCRs, from 5.69 to 2.28 for the Kitchen Package and from 6.82 to 2.15 for the Kohl Package.

Fifth, we halve the cookware and kohl BLL-reduction parameters while leaving the calcium parameter unchanged. This directly tests how much of the result depends on optimistic assumptions about safe-product replacement. The median total BCR falls from 5.69 to 2.98 for the Kitchen Package and from 6.82 to 3.41 for the Kohl Package.

Finally, local kohl practice matters. The main analysis assumes first-year kohl exposure can occur among boys and girls. If infant substitution is restricted to girls while maternal kohl substitution remains unchanged, the median cognition-only BCR is about 38% lower and the median total BCR is about 32% lower than in the all-infants main case. At the same time, in our simulation the fixed costs of the sentinel package are a large share of the total cost. If kohl were distributed alongside other commodities (e.g., as an addition to the kitchen package) its costs would be cut almost in half, which almost doubles its cost-effectiveness. Appendix Table A2 reports the median, 5th percentile, 95th percentile and share of draws below parity for the main robustness checks.

Discussion

Interpretation and implementation

The model suggests a socially cost-effective case for the Kitchen Package and a more assumption-sensitive case for the Kohl Package in targeted settings where hazardous products are common. That result should be read as a demonstration rather than a universal claim. Using plausible LMIC economic values and plausible hotspot parameters, free or heavily subsidised substitution can be socially cost-effective. Just as important, the paper provides a tool for asking whether that remains true in a particular district, region or country once local prevalence, prices, delivery capacity and measured blood lead response are used instead.

The factors that matter most for local decisions are straightforward: how common the hazardous product is, whether the relevant health contact is reached, whether facilities and merchants have the safe item in stock when the household presents, whether caregivers redeem and use the safer product, whether unsafe-product use falls enough to change exposure materially, and how large the resulting blood lead reduction is. In short, the central question for a ministry or NGO is not whether the global model is exactly right, but whether local conditions make substitution attractive in the setting under consideration.

The main practical implication is that implementation quality matters enormously. The pot pathway can fail at several stages: voucher issuance, merchant stock, redemption or residual continued use of unsafe cookware. At the same time, not every step should be treated pessimistically. Conditional on redeeming a voucher for a valuable durable good, use by the pregnant woman or toddler may be high because redemption itself is effortful and self-targeting.

The robustness checks show that this optimism is not costless. When bad news arrives in correlated bundles across local income, ANC reach and implementation quality, or when several cookware and kohl effect-size assumptions deteriorate together, the estimated returns fall substantially. Even so, the median total BCR remains above parity in these deliberately adverse multi-parameter scenarios, suggesting that the basic social case is weakened rather than eliminated.

An important goal of this analysis is to make the assumptions, code and uncertainty structure explicit enough that researchers and policymakers in affected LMIC settings can adapt, challenge and improve the model using local data. See the Data availability section for these resources. The spreadsheet is deliberately simpler than the Python model: it allows a ministry, NGO or research team to change the modal assumptions and quickly see whether the substitution package remains attractive in its own region. The Python code remains the full probabilistic model for users who want to change ranges, correlations or trial-level outputs. Exposure sources, cultural practices such as kohl use, safe-product markets and implementation bottlenecks vary sharply across settings. Thus, any pilot or implementation effort should be led with local investigators, health-system actors and affected communities.

Experience in Adjara, Georgia also suggests that ANC platforms can support lead-related biomonitoring, counselling and follow-up.³⁶ The Sentinel Firewall addresses a different margin. Instead of relying on universal individual testing and then providing mitigation only after an elevated blood lead result, it considers universal safe-product provision to the relevant target group within sentinel-identified high-risk areas. Test-triggered provision can concentrate resources on households with measured high BLL, but it adds testing and follow-up costs and may miss households whose product-related exposure is current or likely but not yet reflected in the measured biomarker. Sentinel-area provision is less individually targeted, but may better protect critical developmental windows when hazardous products are widespread and safe substitutes are inexpensive. In practice,

hybrid strategies may be best, such as identifying high-risk districts through sentinel surveillance and then combining rapid provision with selective biomonitoring. Appendix section 13 formalises this trade-off.

Early pilots can start in one subdistrict or sentinel-clinic catchment, deploying one or two product categories, and measuring uptake and sustained use. Three design details are especially useful for learning. First, a removal bonus could offer a small supplemental good or incentive when households surrender old hazardous products; this bonus would reduce residual exposure and turn surrendered products into enforcement intelligence about suppliers and product chains. Second, community health workers could conduct randomised spot-checks during neonatal or early-childhood home visits to observe whether safer products are being used for the pregnant woman or young child. Third, sentinel biomonitoring at a later infant visit, such as the 9-month immunisation contact where feasible, could verify whether product substitution is reducing blood lead.

Limitations

Several limitations remain. Some blood lead reduction parameters, especially for utensils and kohl, remain informed extrapolations rather than tightly estimated causal effects. Maternal and neonatal pathways use reduced-form DALY multipliers rather than a full disease-course model. The positive dependence structure across implementation steps is stylised and may not fully capture the correlation structure of strong and weak delivery environments. Kohl use varies by sex, age and local practice. Finally, the model does not yet incorporate household-level source correlation or a full source-apportionment model of baseline blood lead.

Conclusion

The model suggests that a programme to replace lead-containing goods with safe pots, utensils, calcium and kohl can generate positive social returns under a wide range of assumptions, if the substitution is targeted to regions with high prevalence of the unsafe products. Equally important, the analysis provides a reusable decision tool. The central question for a ministry or NGO is not whether these exact illustrative parameters apply everywhere, but whether local prevalence, prices, delivery capacity and measured blood lead response make substitution attractive in the specific region under consideration.

At policy scale, the magnitudes are also meaningful: in the deterministic modal run, 1 million annual births in sentinel-identified hotspot districts would yield about US\$68.7 million in discounted net social benefits for the Kitchen Package and about US\$37.1 million for the Kohl Package.

The case for the Sentinel Firewall is not that it replaces regulation, but that it offers governments a practical way to reduce harm immediately while building the capacity and enforcement systems needed to reduce sources of lead exposure. Demand-side substitution through maternal and child health platforms is an emergency, transitional response to a persistent and often informal exposure problem.

End matter

Data availability statement

The code, spreadsheet, and supplementary materials are archived on Zenodo at doi:10.5281/zenodo.19898684 and mirrored in a public GitHub repository at <https://github.com/levine63/Sentinel-Firewall>. The full Python simulation, including Monte Carlo uncertainty, parameter correlations, plots and trial-level exports, is available in the public repository together with a deterministic spreadsheet with editable modal parameter values and live formulas. The spreadsheet is designed for easier local adaptation; the Python code remains the authoritative source for the probabilistic sensitivity analysis.

Ethics statements

Patient consent for publication

Not applicable.

Ethics approval

Not applicable. This is an economic modelling study using published aggregate data and programmatic assumptions; it did not involve human participants or identifiable individual-level data.

Contributors

David I Levine conceived the Sentinel Firewall framework, developed the model structure and assumptions, interpreted the simulation results, drafted and revised the manuscript, and is responsible for the overall content as guarantor.

Author Reflexivity Statement

This article is a conceptual and economic modelling analysis based entirely on secondary literature, published aggregate data and transparent parameter assumptions. It did not involve primary data collection, human participants, local research infrastructure or unpublished data from any LMIC setting. The work was conceived, modelled and written by a sole author based in a high-income country, which is an important limitation for a paper addressing policy choices in LMICs. To reduce the risk of over-generalisation, the manuscript frames the model as a decision tool rather than a setting-specific implementation plan, reports wide uncertainty ranges, provides the code and spreadsheet for adaptation, and emphasises that any pilot or implementation should be designed and led with researchers, health-system actors and affected communities in the relevant country or region.

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Competing interests

None declared.

Patient and public involvement

Patients and the public were not involved in the design, conduct, reporting or dissemination plans for this modelling study. No patients, frontline health workers, ministries or community members were directly engaged in shaping the present model, so there are no stakeholder-engagement effects to report for the analysis itself.

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Figure legends

Figure 1. Two-panel overview of the Sentinel Firewall logic. Panel A shows the core causal pathway from safe-product substitution to lower maternal and child blood lead, improved child neurodevelopment and IQ, and higher lifetime earnings. Calcium and maternal kohl primarily affect the maternal pathway; child utensils and infant kohl primarily affect the child pathway; and safe pots can affect both. Panel B shows the implementation chain linking targeted households to realized population benefit, illustrating why program effects are smaller than the full biological effect in households that switch completely to the safe product. Source file: [figure_1_kitchen_cognition_pathway.png](#).

Figure 2. Probabilistic sensitivity analysis of the Kitchen and Kohl packages. Distribution of cognition-channel benefit-cost ratios from 10 000 Monte Carlo draws, with dashed lines marking parity and the package medians. Source file: [figure_2_bcr_distribution.png](#).

Figure A1. Online appendix only. Probabilistic sensitivity analysis of the Kitchen Package. Distribution of cognition-channel benefit-cost ratios from 10 000 Monte Carlo draws, with dashed lines at parity and the median. Source file: [figure_a1_kitchen_bcr_distribution.png](#).

Figure A2. Online appendix only. Probabilistic sensitivity analysis of the Kohl Package. Distribution of cognition-channel benefit-cost ratios from 10 000 Monte Carlo draws, with dashed lines at parity and the median. Source file: [figure_a2_kohl_bcr_distribution.png](#).

Figure A3. Online appendix only. Tornado diagram for Kitchen Package cognition-only benefit-cost ratio. Source file: [figure_a3_kitchen_tornado.png](#).

Figure A4. Online appendix only. Tornado diagram for Kohl Package cognition-only benefit-cost ratio. Source file: `figure_a4_kohl_tornado.png`.

Figure A5. Online appendix only. Simplified timing pathways for the Kitchen Package. Calcium supplementation affects child exposure only through reduced maternal blood lead during pregnancy. Safe pot voucher and substitution affect child exposure through prenatal transfer, maternal exposure while nursing, and direct food pathways after complementary feeding begins. Safe child utensils affect child exposure only after complementary feeding begins. Source file: `figure_a5_kitchen_timing_pathways.png`.

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